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Impedance matching circuit and method

Abstract

An impedance matching circuit is provided. The impedance matching circuit includes a reference voltage generator configured to generate a reference voltage. A code generator is configured to generate a first calibration code by comparing the reference voltage with a first voltage associated with a first node and a second calibration code by comparing the reference voltage with a second voltage associated with a second node. A first resistance unit is configured to supply the first voltage to the first node in response to the first calibration code to calibrate its resistance to be equal to a reference resistance. A second resistance unit is configured to supply the second voltage to the second node in response to the second calibration code to thereby calibrate its resistance to the reference resistance.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application a continuation of U.S. patent application Ser. No. 17/742,803 titled “Impedance Matching Circuit And Method” and filed May 12, 2022, the disclosure of which is incorporated herein by reference in its entirety.

BACKGROUND

(1) A variety of semiconductor devices embodied by an integrated circuit chip, such as a Central Processing Unit (CPU), a memory and a gate array, have been combine in electrical products such as a personal computer, a server, and a work station. In most cases, the semiconductor devices with a receiving circuit for receiving signals from the outside through input pads and an output circuit for providing internal signals to the outside through output pads.

(2) The input/output pads are connected to a transmission path. Electrical signals are reflected back when the impedance differs. The impedance mismatching is caused by variation of a manufacture

process, a supply voltage and an operation temperature (PVT). The impedance mismatching makes it hard to transmit data at high speed the signals are distorted by the impedance mismatching.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.
- (2) FIG. 1 is a block diagram showing an impedance matching circuit in accordance with some embodiments.
- (3) FIG. 2 is a block diagram showing a Process, Voltage and Temperature (PVT) calibration circuit, in accordance with some embodiments.
- (4) FIG. 3A is a graph illustrating an output voltage of a comparator for a first post driver calibration, in accordance with some embodiments.
- (5) FIG. 3B is a graph illustrating an output voltage of a comparator for a second post driver calibration, in accordance with some embodiments.
- (6) FIG. 4A illustrates an equivalent circuit diagram for a first post driver calibration, in accordance with some embodiments.
- (7) FIG. 4B illustrates an equivalent circuit diagram for a second post driver calibration, in accordance with some embodiments.
- (8) FIG. 5 is a graph illustrating calibration voltages in accordance with some embodiments.
- (9) FIG. 6 is a flow diagram illustrating a method of impedance matching, in accordance with some embodiments.

DETAILED DESCRIPTION

- (10) The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.
- (11) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.
- (12) In order to reduce the impact of environmental variation, an impedance matching circuit is described. The disclosed impedance matching circuit can be embedded in a semiconductor device, for example, a memory device or a processing device. By embedding the impedance matching circuit, the performance of input/output interfaces of the semiconductor device will be independent of a Process, Voltage, and Temperature (PVT) variation. In examples, the disclosed impedance

matching circuit uses only one comparator which leads to decreased power consumption and increased accuracy.

(13) FIG. 1 is a block diagram showing an impedance matching circuit **100** in accordance with some embodiments. As shown in FIG. 1, impedance matching circuit **100** includes a PVT calibration circuit **102**. PVT calibration circuit **102** includes a replica post driver **104** (also referred to as a replica resistance unit), a pad **106** (also referred to as a first node), a comparator **108**, and a counter **110**. Impedance matching circuit **100** further includes an external resistor **112** (also referred to as a reference resistor) and a post driver **114** (also referred to as resistance unit).

(14) Replica post driver **104** is connected to pad **106**. A first terminal of external resistor **112** is connected to pad **106** and a second terminal of external resistor **112** is connected to the ground. A first input terminal of comparator **108** is connected to pad **106** and a second input terminal of comparator **108** is connected to a reference voltage node which provides a reference voltage. An output terminal of comparator **108** is connected to a counter **110**. Power for post driver **114** is provided from a voltage VDDQ. Power for replica post driver **104** is provided through a supply voltage VDD and the voltage VDDQ. In examples, the supply voltage is approximately equal to 150 mV and the voltage VDDQ is approximately equal to 300 mV. However, other voltages are possible. In addition, the reference voltage is approximately equal to one half of the voltage VDDQ (that is, 150 mV). Moreover, in examples, a resistance value of external resistor **112** is 50 ohms. However, other resistance values are possible.

(15) For PVT calibration, comparator **108** compares a pad voltage (also referred to as a first voltage) received at the first input terminal with the reference voltage received at the second input terminal and provides an output signal based on the comparison on the output terminal. The output signal of comparator includes an UP/DOWN signal. Counter **110** receives the UP/DOWN signal from the output terminal of comparator **108** and generates a binary code, also referred to as a PUCODE<0:N> (for example, PUCODE<0>, PUCODE<0>, . . . , PUCODE<n>). The PUCODE<0:N>, also referred to as calibration codes, is provided to both replica post driver **104** and post driver **114**. PUCODE<0:N> is used to calibrate a resistance value of replica post driver **104** to match that of external resistor **112**. For example, the PUCODE<0:N> is used to turn ON or turn OFF one or more transistors in replica post driver **104** to calibrate a resistance value of replica post driver **104** to match that of external resistor **113**. Similarly, the PUCODE<0:N> is used to calibrate a resistance value of post driver **114** to match that of external resistor **112**. For examples, the PUCODE<0:N> is used to turn ON or turn OFF one or more transistors in post driver **114** to calibrate a resistance value of post driver **114** to match that of external resistor **112**.

(16) FIG. 2 is block diagram showing PVT calibration circuit **102** in accordance with some embodiments. As shown in FIG. 2, PVT calibration circuit **102** includes a first post driver **202** and a second post driver **204**. First post driver **202** is also referred to as a pull-up post driver or a first resistance unit and second post driver **204** is also referred to as a pull-down post driver or a second resistance unit. In some examples, each of first post driver **202** and second post driver **204** include a plurality of resistors and a plurality of transistors. Each of the plurality of resistors of first post driver **202** are connected to one of the plurality of transistors at a first internal node in series. In addition, each of the plurality of resistors of first post driver **202** are connected in parallel to each other. Similarly, each of the plurality of resistors of second post driver **204** are connected to one of the plurality of transistors at a second internal node in series. In addition, each of the plurality of resistors of second post driver **204** are connected in parallel to each other.

(17) PVT calibration circuit **102** further includes a reference voltage generator **206**, a first data selector **208** (also referred to as a first multiplexer **208**), a second data selector **210** (also referred to as a second multiplexer **210**), an offset voltage **212**, and an inverter **214**. In addition, PVT calibration circuit **102** includes comparator **108** and counter **110**. In examples, both first multiplexer **208** and second multiplexer **210** are 2:1 multiplexers. In examples, comparator **108** is an operation amplifier. In some examples, reference voltage generator **206** includes a plurality of

resistors connected in series between a voltage VDDQ node and the ground. In examples, first multiplexer **208**, comparator **108**, second multiplexer **210**, inverter **214**, and counter **110** form a code generator circuit.

(18) As shown in FIG. 2, first post driver **202** is connected between a voltage VDDQ node and pad **106**. External resistor **112** is connected between pad **106** and the ground. A first input terminal of first multiplexer **208** is connected to pad **106**. A second input terminal of first multiplexer **208** is connected to second post driver **204** at a second node **216**. Thus, second post driver **204** is connected between a second node **216** and the ground. In examples, first post driver **202** provides a first voltage at pad **106** and second post driver **204** provides a second voltage at a second node **214**.

(19) An output terminal of first multiplexer **208** is connected to a first input terminal of comparator **108**. A second input terminal of comparator **108** is connected to reference voltage generator **206** through offset voltage **212**. An output terminal of comparator **108** is connected to a first input terminal of second multiplexer **210** and an input terminal of inverter **214**. An output terminal of inverter **214** is connected to a second input terminal of second multiplexer **210**. An output terminal of second multiplexer **210** is connected to counter **110**. An output terminal of counter **110** is connected to both first post driver **202** and second post driver **204**.

(20) During the calibration operation, first multiplexer **208** chooses one of first post driver **202** and second post driver **204** to be calibrated. In some examples, first multiplexer **208** chooses first post driver **202** to be calibrated first and then chooses second post driver **204** to be calibrated after the calibration of first post driver **202**. During the calibration of first post driver **202** (also referred to as a pull-up calibration) the first voltage at pad **106** becomes the same or nearly the same as the reference voltage. When first post driver **202** is chosen for calibration, the first input terminal of comparator **108** is connected to pad **106**. Comparator **108** thus receives the pad voltage (that is, the first voltage) at the first input terminal and the reference voltage offset by the offset voltage V_{os} at the second input terminal. Comparator **108** then compares the pad voltage with the reference voltage and provides an output signal at the output terminal based on the comparison. Inverter **214** receives the output signal from comparator **108**, inverts the output signal, and provides the inverted output signal at the output terminal.

(21) Thus, the first input terminal of second multiplexer **210** receives the output signal as a first input from comparator **108** and the second input terminal of second multiplexer **210** receives the inverted output signal as a second input from inverter **214**. Second multiplexer **210** selected one of the two inputs and provides the selected input to counter **110**. In some examples, for calibrating first post driver **202**, the inverted output signal is selected. Counter **110** generates the PUCODE<0:N> (also referred to as first calibration codes) for first post driver **202** based on the received output signal. The PUCODE <0:N> is provided to first post driver **202** that adjust its resistance to match that of external resistor **112**. For examples, the PUCODE<0:N> is used to turn ON or turn OFF one or more transistors in first post driver **202** to calibrate its resistance to match that of external resistor **112**. The calibrated resistance of first post driver **202** effects the first voltage of pad **106**, and this operation is repeated. As a result, the PUCODE <0:N> is counted until the resistance value of first post driver **202** becomes identical or nearly identical to the resistance value of external resistor **112**.

(22) Like the pull-up calibration, a pull-down calibration (that is the calibration of second post driver **204**) starts in a manner that the second voltage at second node **216** becomes the same or nearly the same as the reference voltage. In other words, the pull-down calibration is performed such that the resistance value of second post driver **204** becomes the same or nearly the same as that of external resistor **112**. For example, after calibration of first post driver **202**, first multiplexer **208** selects second post driver **204** for calibration. First multiplexer **208** provides a second voltage of second node **216** at the second input terminal as an output to comparator **108**. Comparator **108** compares the second voltage received at the first input terminal with the reference voltage offset by the offset voltage V_{os} received at the second input terminal. Comparator **108** provides an output

signal at the output terminal based on the comparison. Inverter **214** receives the output signal from comparator **108**, inverts the output signal and provides the inverted output signal at the output terminal.

(23) Thus, the first input terminal of second multiplexer **210** receives the output signal as a first input from comparator **108** and the second input terminal of second multiplexer **210** receives the inverted output signal as a second input from inverter **214**. In some examples, for calibrating second post driver **204**, the non-inverted output signal is selected. Second multiplexer **210** selected one of the two inputs and provides the selected input to counter **110**. Counter **110** generates the PDCODE<0:N> (also referred to as a second calibration code) for second post driver **204** based on the received output signal. The PDCODE <0:N> is provided to second post driver **204** that adjust its resistance to match that of external resistor **112**. For examples, the PDCODE<0:N> is used to turn ON or turn OFF one or more transistors in second post driver **204** to calibrate its resistance to match that of external resistor **112**. In some examples, for calibrating second post driver **204**, both PDCODE<0:N> and PUCODE<0:N> codes are used. For examples, first portion of second post driver **204** is calibrated using the PUCODE<0:N> and then a second portion of second post driver **204** is calibrated using the PDCODE<0:N>.

(24) FIG. 3A is a graph illustrating the output signal of comparator **108** for first post driver **202** calibration. For examples, first plot **310** of FIG. 3A illustrates the output signal of comparator **108** for first post driver **202** calibration. As shown in FIG. 3A, the output signal is a logic high (shown as VOH) until the first voltage (that is, the pad voltage) is less than the reference voltage plus the offset voltage Vos. When, the first voltage (that is, the pad voltage) is greater than the reference voltage plus the offset voltage Vos, the output signal of comparator **108** drops to a logic low (shown as VOL).

(25) FIG. 3B is a graph illustrating the output signal of comparator **108** for second post driver **204** calibration. For examples, second plot **320** of FIG. 3B illustrates the output signal of comparator **108** for second post driver **204** calibration. As shown in FIG. 3B, the output signal is a logic high (shown as VOH) until the second voltage (that is, the pulldown voltage) is less than the reference voltage plus the offset voltage Vos. When, the second voltage is greater than the reference voltage plus the offset voltage Vos, the output signal of comparator **108** drops to a logic low (shown as VOL). Thus, and as shown in FIGS. 3A and 3B, the effect of the offset voltage is predictable and in a same fashion for both first post driver **202** calibration and second post driver **204** calibration. This results in better calibration of both first post driver **202** calibration and second post driver **204** calibration.

(26) FIG. 4A illustrates an equivalent circuit diagram for first post driver **202** calibration. As shown in FIG. 4A, the equivalent circuit diagram for first post driver **202** calibration includes a first post driver resistor Rpu **410** connected in series with external resistor **112**. For example, a first terminal of first post driver resistor Rpu **410** is connected to the voltage VDDQ node and a second terminal of first post driver resistor Rpu **410** is connected to pad **106**. External resistor **112** is connected between pad **106** and the ground. In examples, a resistance value of external resistor **112** is approximately equal to 50 Ohms and the voltage VDDQ is approximately equal to 300 mV. Hence, the voltage of pad **106**, which is one of half of the voltage VDDQ offset by the offset voltage Vos is approximately equal to 155 mv (that is, 150 mV+5 mV). Therefore, the resistance value of first post driver resistor Rpu **410** is determined as:

$$(27) \frac{50}{50 + R_{pu}} \times 300 = 150 + 5$$

Rpu=46.77. The resistance value of first post driver resistor Rpu **410** is therefore offset by 3.23 ohms from that of external resistor **112** which has a resistance value of 50 ohms.

(28) FIG. 4B illustrates an equivalent circuit diagram for second post driver **204** calibration. As shown in FIG. 4B, the equivalent circuit diagram for second post driver **204** calibration includes first post driver resistor Rpu **410** connected in series with a second post driver resistor Rpd **420**. For example, the first terminal of first post driver resistance Rpu **410** is connected to the voltage

VDDQ node and the second terminal of post driver resistance Rpu **410** is connected to second node **216**. Therefore, the resistance value of second post driver **204**, that is, second post driver resistance Rpd **420** is determined as:

$$(29) \frac{R_{pd}}{46.77 + R_{pd}} \times 300 = 150 + 5$$

Rpd=49.99. The resistance value of second post driver resistor Rpd **420** is therefore offset by 0.01 ohms from that of external resistor **112** which has a resistance value of 50 ohms. Hence, the deviation in second post driver resistor Rpd **420** is less than the deviation in first post driver resistor Rpd **410**. Thus, the accuracy of the calibration is increased.

(30) FIG. 5 is a graph illustrating calibration voltages in accordance with some embodiments. For examples, first plot **510** illustrates the first voltage, second plot **520** illustrates second voltage, and third plot **530** illustrates the reference voltage. As shown in FIG. 5, the first voltage converges on the reference voltage.

(31) FIG. 6 illustrates a flow diagram of a method **600** for impedance matching, in accordance with some embodiments. In examples, method **600** can be practiced in impedance matching circuits described with reference to FIGS. 1-2 above. In some examples, method **600** can be stored as instructions on a storage medium and when executed by a processor connected to the storage medium perform method **600**.

(32) At block **610** of method **600**, the reference voltage is generated. For examples, reference voltage generator **206** generates the reference voltage. In some examples, the reference voltage is half of the voltage VDDQ.

(33) At block **620** of method **600**, a first calibration code is generated based on comparing the reference voltage with a first voltage associated with a first node. For example, comparator **108** compares the first voltage with the reference voltage and generates the output signal based on the comparison. Counter **110** then generates the first calibration code (that is, the PUCODE<0:N>) from the output signal.

(34) At block **630** of method **600**, a resistance value of a first resistance unit is calibrated based on the first calibration code to be equal to a reference resistance. The first resistance unit is connected to the first node. For example, the resistance value of first post driver **202** is calibrated based on the first calibration code to be equal to the resistance value of external resistor **112**.

(35) At block **640** of method **600**, a second calibration code is generated based on comparing the reference voltage with a second voltage associated with a second node. For example, comparator **108** compares the second voltage with the reference voltage and generates the output signal based on the comparison. Counter **110** then generates the second calibration code (that is, the PDCODE<0:N>) from the output signal.

(36) At block **650** of method **600**, a resistance value of a second resistance unit is calibrated based on the second calibration code to be equal to the reference resistance. The second resistance unit is connected to the second node. For example, the resistance value of second post driver **204** is calibrated based on the second calibration code to be equal to the resistance value of external resistor **112**.

(37) In accordance with example embodiments, an impedance matching circuit comprises: a reference voltage generator configured to generate a reference voltage; a code generator configured to generate a first calibration code by comparing the reference voltage with a first voltage associated with a first node and a second calibration code by comparing the reference voltage with a second voltage associated with a second node; a first resistance unit configured to supply the first voltage to the first node in response to the first calibration code to calibrate its resistance to be substantially equal to a reference resistance; and a second resistance unit configured to supply the second voltage to the second node in response to the second calibration code to calibrate its resistance to the reference resistance.

(38) In example embodiments, an impedance matching circuit comprises: an external resistor connected between a first node and ground; a first resistance unit connected to the first node; a

second resistance unit connected between a second node and the ground; a reference voltage generator configured to generate a reference voltage; a code generator configured to generate a first calibration code by comparing the reference voltage with a first voltage associated with the first node and a second calibration code by comparing the reference voltage with a second voltage associated with the second node, wherein: the first resistance unit, in response to the first calibration code, is configured to calibrate its resistance to be substantially equal to a reference resistance; and the second resistance unit, in response to the second calibration code, is configured to calibrate its resistance to be substantially equal to the reference resistance.

(39) In accordance with example embodiments, a method for calibrating an impedance of a device, comprises: generating a reference voltage; generating a first calibration code based on comparing the reference voltage with a first voltage associated with a first node; calibrating, based on the first calibration code a resistance value of a first resistance unit to be substantially equal to a reference resistance, wherein the first resistance unit is connected to the first node; generating a second calibration code based on comparing the reference voltage with a second voltage associated with a second node; and calibrating, based on the second calibration code, a resistance value of a second resistance unit to be substantially equal to the reference resistance, wherein the second resistance unit is connected to the second node.

(40) The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. An impedance matching circuit comprising: a reference voltage generator configured to generate a reference voltage; a code generator comprising a first multiplexer and a comparator, wherein the first multiplexer selects one of a first voltage associated with a first node and a second voltage associated with a second node and provides the selected voltage to the comparator, wherein the comparator is configured to generate one of a first calibration code by comparing the reference voltage with the first voltage and a second calibration code by comparing the reference voltage with the second voltage; a first resistance unit configured, in response to the first calibration code, to calibrate its resistance to be substantially equal to a reference resistance; and a second resistance unit configured, in response to the second calibration code, to calibrate its resistance to the reference resistance.

2. The impedance matching circuit of claim 1, wherein the first multiplexer is configured to select the first voltage to calibrate the first resistance unit, and wherein the comparator is configured to compare the first voltage with the reference voltage received at a second input terminal of the comparator.

3. The impedance matching circuit of claim 2, wherein the code generator further comprises a second multiplexer and an inverter, wherein the comparator is configured to provide an output signal to a first input terminal of the second multiplexer and to an input terminal of the inverter based on comparing the first voltage with the reference voltage.

4. The impedance matching circuit of claim 3, wherein the inverter is configured to invert the output signal received from the second multiplexer and to provide an inverted output signal to a second input terminal of the second multiplexer, wherein the second multiplexer is configured to select one of the output signal received at the first input terminal of the second multiplexer and the

inverted output signal received at the second input terminal of the second multiplexer.

5. The impedance matching circuit of claim 4, wherein the code generator further comprises a counter, wherein the counter is configured to receive the selected output signal from the second multiplexer and to generate the first calibration code based on the received selected output signal.
6. The impedance matching circuit of claim 1, wherein the first multiplexer is configured to select the second voltage to calibrate the second resistance unit, and wherein the comparator is configured to compare the second voltage with the reference voltage received at a second input terminal of the comparator.
7. The impedance matching circuit of claim 6, wherein the code generator further comprises a second multiplexer and an inverter, wherein the comparator is configured to provide an output signal to a first input terminal of the second multiplexer and to an input terminal of the inverter based on comparing the second voltage with the reference voltage.
8. The impedance matching circuit of claim 7, wherein the inverter is configured to invert the output signal received from the second multiplexer and to provide an inverted output signal to a second input terminal of the second multiplexer, and wherein the second multiplexer is configured to select one of the output signal received at the first input terminal of the second multiplexer and the inverted output signal received at the second input terminal of the second multiplexer.
9. The impedance matching circuit of claim 8, wherein the code generator further comprises a counter, wherein the counter is configured to receive the selected output signal from the second multiplexer and to generate the second calibration codes based on the received selected output signal.
10. An impedance matching circuit comprising: an external resistor connected between a first node and ground; a first resistance unit connected to the first node; a second resistance unit connected between a second node and the ground; a reference voltage generator configured to generate a reference voltage; and a code generator comprising a first multiplexer and a comparator, wherein the first multiplexer selects one of the first voltage associated with a first node and the second voltage associated with a second node and provides the selected voltage to the comparator, wherein the comparator is configured to generate a first calibration code by comparing the reference voltage with the first voltage and a second calibration code by comparing the reference voltage with the second voltage, wherein: the first resistance unit, in response to the first calibration code, is configured to calibrate its resistance to be substantially equal to a reference resistance; and the second resistance unit, in response to the second calibration code, is configured to calibrate its resistance to be substantially equal to the reference resistance.
11. The impedance matching circuit of claim 10, wherein the reference voltage is half of a supply voltage.
12. The impedance matching circuit of claim 10, wherein the reference voltage generator is configured to generate the reference voltage by dividing a supply voltage.
13. The impedance matching circuit of claim 10, wherein the reference voltage generator includes a plurality of resistors connected in series between a supply voltage node and ground.
14. The impedance matching circuit of claim 10, wherein the first resistance unit comprises: a plurality of resistors connected in parallel between a first internal node and ground; and a plurality of transistors connected to the plurality of resistors.
15. The impedance matching circuit of claim 14, wherein the plurality of transistors are switched ON or OFF based on the first calibration code to match resistance of first resistance unit to be equal to the reference resistance.
16. The impedance matching circuit of claim 10, wherein the second resistance unit comprises: a plurality of resistors connected in parallel between a second internal node and ground; and a plurality of transistors connected to the plurality of resistors.
17. The impedance matching circuit of claim 16, wherein the plurality of transistors are switched ON or OFF based on the second calibration code to match resistance of the second resistance unit

to be equal to the reference resistance.

18. A method for calibrating an impedance of a device, the method comprising: generating, by a reference generator, a reference voltage; selecting, by a first multiplexer of a code generator, a first voltage associated with a first node from the first voltage and a second voltage associated with a second node; comparing, by a comparator of the code generator, the first voltage with the reference voltage; generating, by the code generator, a first calibration code based on comparing the reference voltage with the first voltage and a second calibration code based on comparing the reference voltage with a second voltage associated with a second node; calibrating, based on the first calibration code, a resistance value of a first resistance unit to be substantially equal to a reference resistance, wherein the first resistance unit is connected to the first node; and calibrating, based on the second calibration code, a resistance value of a second resistance unit to be substantially equal to the reference resistance, wherein the second resistance unit is connected to the second node.

19. The method of claim 18, wherein calibrating, based on the first calibration code, the resistance value of the first resistance unit to be substantially equal to the reference resistance comprises switching ON or OFF one or more transistors in the first resistance unit based on the first calibration code.

20. The method of claim 18, further comprising: selecting, by the first multiplexer of the code generator, the second voltage associated with the second node; comparing, by the comparator of the code generator, the second voltage with the reference voltage; and generating, by the code generator, the second calibration code based on comparing the reference voltage with the second voltage.
